

Assessing human perception

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Network technology typically contains complex and nonlinear processes, such as low bit rate codecs for data reduction. This nonlinearity means that the perceived performance cannot be predicted with conventional engineering performance metrics. Reliable objective assessment of perceived performance is required for optimal design, commissioning and monitoring of quality. This paper looks at how BT can measure, optimise and control the perceived performance of the service that is provided.

The final receiver of a signal is the human user — by modelling the senses of the human it is possible to predict whether an error would be perceptible. An auditory perceptual model has been developed and used to predict the subjective performance of a variety of nonlinear systems. Applications for the auditory model include design optimisation, as a measurement tool and as part of a nonlinear network planning tool.

This paper focuses specifically on the auditory system, and how perceptual modelling is providing the basis for improved quality of voice service. However, as the global landscape adapts over the next decade, with multimedia, personal interactivity, and the concept of 'virtuality' then all of the human senses need to be considered together. Future work is planned to include an elementary visual model and the combination of sensory models for the assessment of multimedia products and services.

1. Introduction

Intense competition and rising customer expectations are ensuring that price differentials are minimised in the telecommunications market-place. When price is desensitised in this way, other factors such as quality become critical differentiators. A key requirement, therefore, is the need for a reliable means of assessing the quality of performance experienced by the customer. Not only must acceptable standards be met consistently, but this must be done at minimum cost to the operator. Objective assessment of parameters, such as speech quality, will enable standards to be efficiently maintained together with effective assessment of systems during design, commissioning and operation.

The network increasingly contains complex and nonlinear processes such as low bit rate coding schemes for data reduction in mobile communications (see Wong et al [1]). Such processes are designed to operate on a speech signal and may deliberately exploit psycho-acoustic phenomena such as masking [2] to minimise the audibility of error artefacts [3]. Conventional engineering performance metrics are inadequate for predicting the subjective performance of such systems where objective assessment requires an analysis which can predict the response of the human senses.

In response to this requirement for a new measurement approach, a technique has been developed which attempts to

model the human senses in order to predict the perceptibility of any errors. Performance prediction based on models of the human senses can be referred to as perceptual analysis. This paper discusses an audio perceptual analysis methods which has been developed at BT Laboratories and indicates the main applications. Future work towards the objective assessment of multimedia products and services is also noted.

2. A new measurement approach — perceptual analysis

Perceptual analysis models the human senses which are the final arbiter of system performance. Because the analysis is independent of the complexity of the distorting process and the distortion characteristic, it is potentially a highly robust measurement approach and can be used to assess the full range of single and multiple nonlinear processes. It is this potential robustness in real world applications, where the distortion type will not generally be known in advance, which makes a sensory modelling approach preferable to a purely empirical approach.

Purely empirical approaches have been described [4—6], where a variety of objective measures such as coherence and cepstral distance are used to train a neural network.

However, the performance of the analysis with an unknown distorting process is uncertain. The performance of a human sensory model approach is potentially independent of distortion type. Perceptual measurement techniques are a rapidly emerging new area and several models based on this approach have appeared in the literature [7–11]. Some of these models, e.g. Paillard B et al [8] and Stuart [11], are intended for the evaluation of hi-fi (music) codecs and are only suitable for predicting the onset of audibility of errors and not the subjectivity of errors which are audible. It is this subjectivity of error that is pertinent to network performance.

The requirements for a perceptual analysis process are a test stimulus which will maximally exercise the device under test, and an analysis which can predict the perceived performance. Requirements for a speech-like test stimulus are discussed in Hollier et al [7] (see Fig 1). The perceptual analysis consists of two parts, a sensory layer and a perceptual layer.

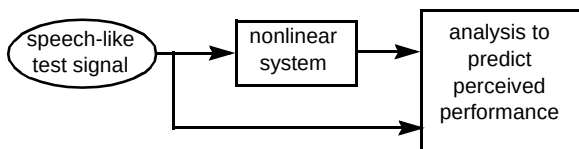


Fig 1 Performance analysis with speech-like test-stimulus.

2.1 Sensory layer

The sensory layer model reproduces the main psycho-physical transformations of human hearing including representation on a critical band or equivalent rectangular bandwidth scale, level to loudness transformation, and an approximation to simultaneous and temporal masking. Reproducing these transformations allows an audio stimulus to be represented with perceptually relevant dimensions. The BT Laboratories analysis represents the perceived audio signal as loudness per unit of auditory bandwidth, i.e. a specific loudness pattern. The subjective loudness of a signal, in some, is then the area under the loudness curve.

Figure 2 shows the application of a subset of these psychoacoustic mappings to calculate the subjective loudness of a 2 kHz tone at a level of -16.6 dBV when heard via a telephone handset.

Original and degraded signals can be compared in this perceptual space in order to predict audible error. Visualisation of audio signals and audible errors in perceptual space can be used as a diagnostic tool. Figures 3 and 4 show the specific loudness pattern and corresponding audible error surface for a fragment of speech processed by a regular-pulse-excited linear-predictive-codec with long-term (pitch) prediction (as used in GSM) and a net bit rate of 13 kbit/s [12].

Careful examination of this audible error surface, with reference to the specific loudness pattern, shows that audible errors occur for:

- voiced sounds — particularly where the rate of change of amplitude is high,
- unvoiced sounds — particularly during onset and transition.

However, in many cases diagnostic interpretation by expert users is not sufficient and a typical requirement is to interpret the audible error surface in order to predict an overall user opinion of performance. This requires the subjectivity of any errors to be algorithmically assessed which is performed by a perceptual layer model.

2.2 Perceptual layer

An important and novel feature of the audio perceptual analysis developed at BT Laboratories is that it is able to automatically interpret the subjectivity of audible errors in order to predict user opinion using a perceptual layer algorithm. This is a key requirement for a measurement system suitable for assessing the speech quality of a wide range of real telecommunications systems.

In order to algorithmically assess the subjectivity of audible errors it is necessary to model the low-level psychoacoustics which determine this subjectivity. It is apparent that the total audible error is not a sufficient indicator of error significance, e.g. a given total error could permeate the entire error surface (perhaps due to a frequency response shaping) or could be collected into one or more discrete features.

Extensive examination of audio subjective test data resulted in the hypothesis that three criteria can be described which determine the subjectivity of audible errors:

- total amount of error — referred to as error-activity,
- distribution of error — referred to as error-entropy,
- correlation of error with original signal — referred to as error-correlation.

Inspiration for error descriptors which describe the first two of these criteria was taken from published work on the adaptive transform coding of video images [13]. Parameters are described which control the dynamic allocation of bits depending on the distribution of energy within the image. The application of these parameters to audible error surface

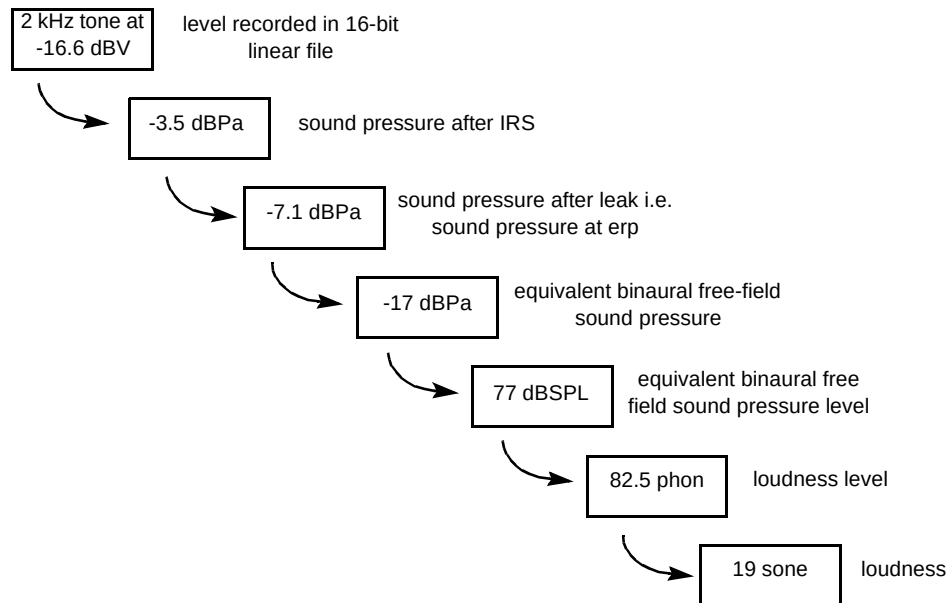


Fig 2 Example calculation of auditory stimulation due to 2 kHz tone.

interpretation is discussed in more detail elsewhere [14—16].

In summary, and with reference to Fig 5, the total audible error may be referred to as the error activity. Based on the expression in Mester and Franke [13], for block activity, error activity can be defined as:

$$E_a = \sum_{i=1}^n \sum_{j=1}^m |e(i, j)| \quad \text{..... (1)}$$

where n and m are the dimensions of the error surface or error-surface fragment and $e(i, j)$ is the audible error in the i th row and j th column. The distribution of the audible-error may be referred to as the error entropy. Again an expression may be based on that given in Mester and Franke [13] to define a suitable quantity:

$$E_e = - \sum_{i=1}^n \sum_{j=1}^m a(i, j) \cdot \ln(a(i, j)) \quad \text{..... (2)}$$

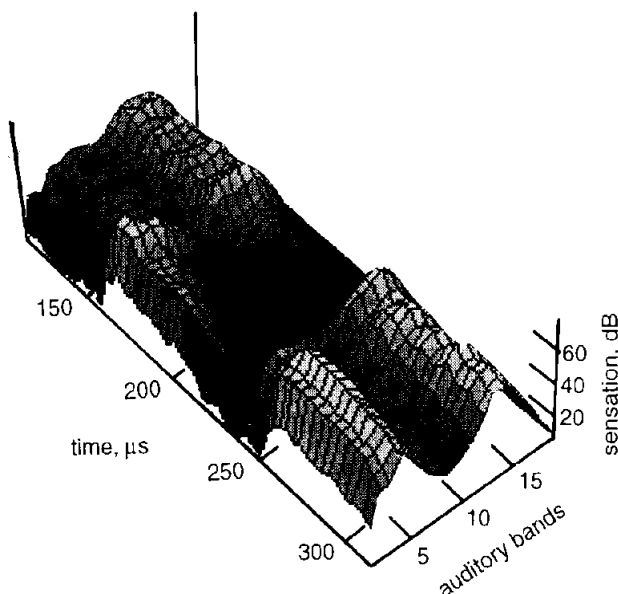


Fig 3 Specific loudness pattern for a speech fragment.

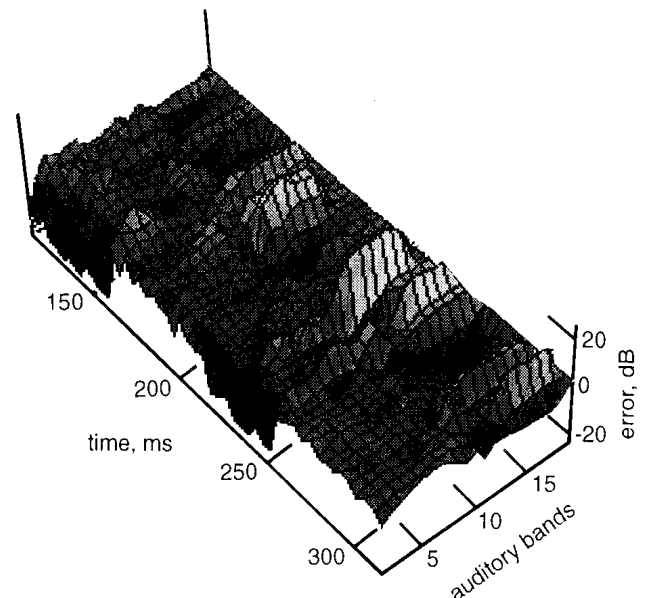


Fig 4 Audible error pattern for speech fragment in Fig 3.

where $a(i, j) = |e(i, j)| / E_a$

The behaviour of the error-activity and error-entropy quantities is readily visualised and is illustrated in Figs 6 to 8, using hypothetical error-surface fragments which show the same error-activity distributed in different ways; these are summarised in Table 1.

Table 1 Error descriptors for hypothetical error-surface fragments.

	Error activity	Error entropy
response shaping (Fig 6)	250	4.63
two peaks (Fig 7)	250	3.78
single peak (Fig 8)	250	0.87

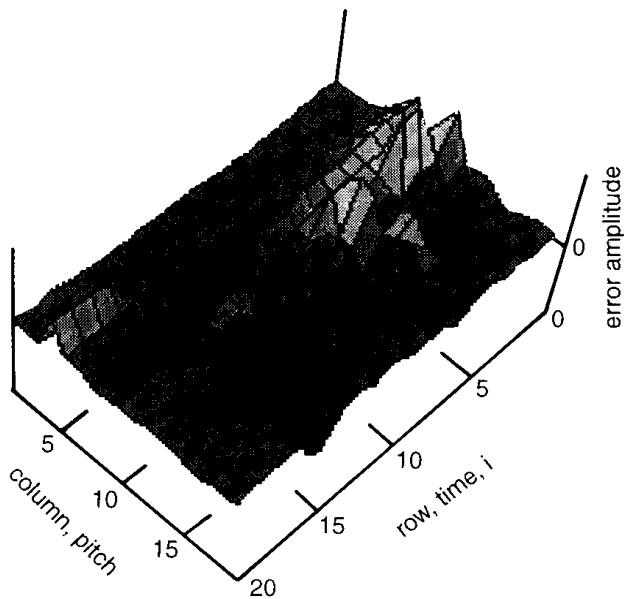


Fig 5 Error-surface fragment.

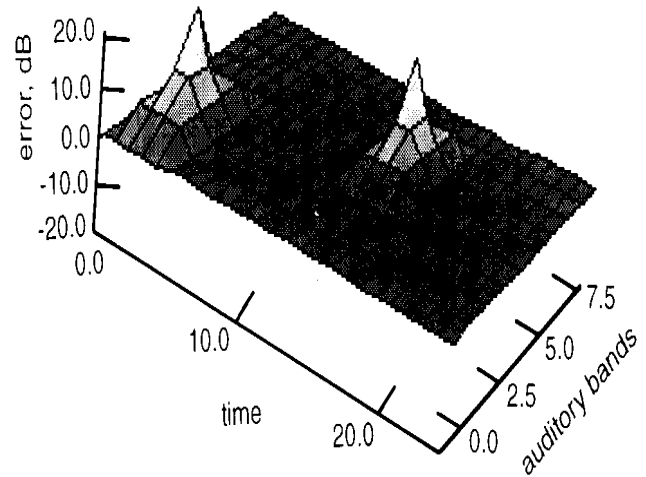


Fig 7 Error-surface fragment — two peaks.

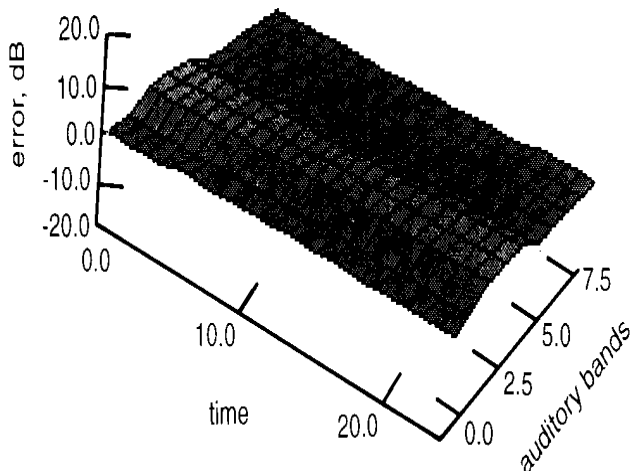


Fig 6 Error-surface fragment — response shaping.

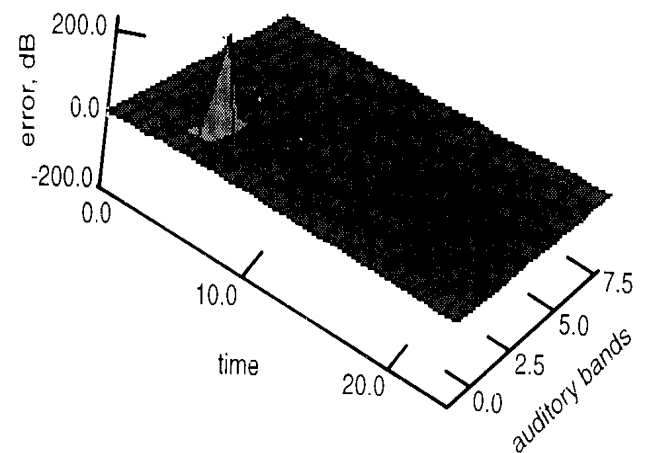


Fig 8 Error-surface fragment — single peak.

Descriptors based on these error parameters can be calculated and their final combination calibrated using the results of a subjective experiment. Predictions by the resulting model can then be compared with further subjective test results. The combination of error descriptors and the prediction performance of the model are reported in Hollier and Hawksford [17].

These error-activity and error-entropy parameters were averaged across the sentence pairs for a subjective experiment and fitted to the subjective test data using a sigmoid function. The resulting combination equation is shown below and can be used to predict the subjective test results from further subjective experiments. The error parameters and opinion score predictions from an international subjective test to select the new half-rate GSM codec [18, 19] are shown in Figs 9 and 10.

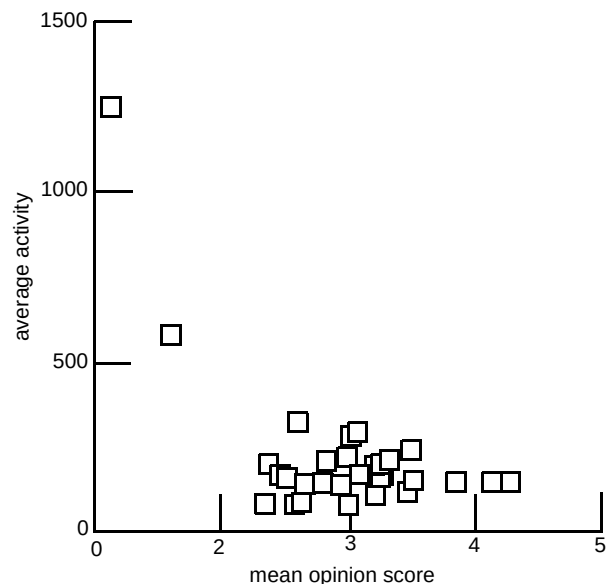
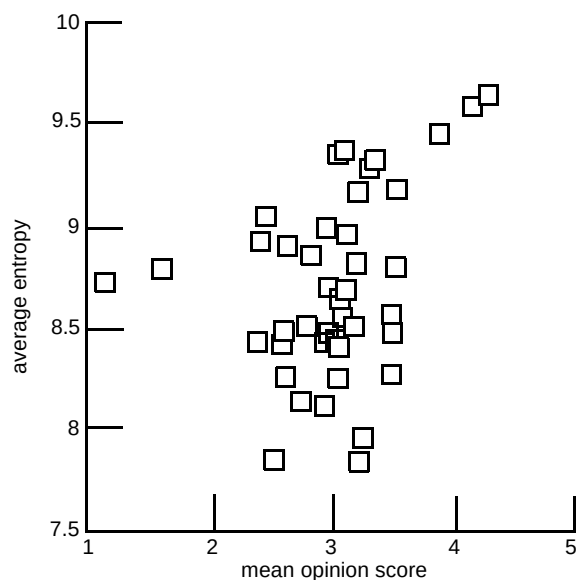


Fig 9 Error parameters.

Table 2 Average error and predicted MOS for different degradations.

Degradation	Subjective ranking	Average Log ₁₀ (error)	MOS prediction
DCME	Best 1	7.288	4.04
Speech codec A (EPX)	2	7.232	3.12
Nonlinear amplifier	3	7.810	2.97
Speech codec C (EPY)	Worst 4	7.555	2.36

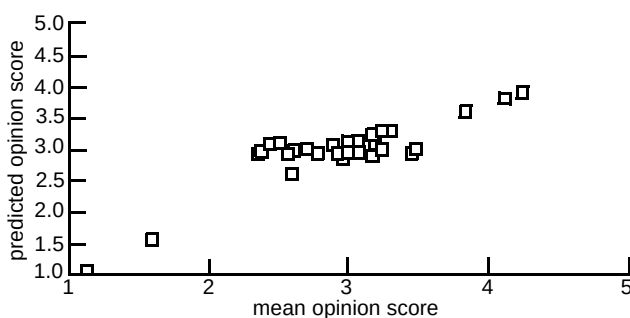


Fig 10 Predicted opinion score.

The combination function was:

$$\begin{aligned} \text{logit } Y = & -132.18342 + 0.00307181 * E_a \\ & -0.00001425 * (E_a)^2 + 50.7525376 * E_e \\ & -6.4591057 * (E_e)^2 + 0.27230122 * (E_e)^3 = w \end{aligned}$$

where:

$$Y = (\text{MOS} - 1)/4, \text{ and} \\ \text{logit } Y = \ln [Y / (1 - Y)]$$

The predicted score MOS' is then given by:

$$\text{MOS}' = 4 * Y' + 1,$$

where $Y' = [\exp(w) / (1 + \exp(w))]$, as described in Hollier and Hawksford [14].

The resulting function gives a mean squared error of 0.11 of an opinion score, with a correlation coefficient of 0.92. The prediction performance has also been tested across a variety of nonlinear distortions and example results are shown in Table 2.

It is apparent that the average error (total error averaged across test sentences) does not indicate the correct subjective ranking, while the MOS prediction, which takes account of the distribution of the error, does predict the correct subjective ranking.

3. Higher level perceptual considerations

The psychoacoustics modelled by the perceptual layer are low level, in terms of their neurological order, and describe the basic subjectivity of audible errors. These low-level properties do not account for higher level psychology and cognition which also contribute to the formation of subjective opinion. For example, the low-level psychoacoustic model may predict a particular degree of error subjectivity, but this may be strongly biased by the degree of attention paid to that sensory modality. Further, subjective opinion expressed by populations of users are dependent on the expectation of those users in connection with a particular product, service or activity. Aspects of speech cognition are

also significant — in particular utterance onsets are generally more critical for speech recognition than the relative steady-state portions of speech sounds.

3.1 *Expectation factor*

The actual opinions expressed by a group of subjects for a given audio quality will vary according to both the quantity to be judged (e.g. intelligibility versus quality) and the expectations of that user group. The quantity judged, e.g. Y_{LE} listening effort, is accounted for directly in the subjective data used to calibrate the analysis predictions. However, user expectation has also been found to be significant. An example of this was apparent during the course of this research — namely the variation in expectation between users of fixed, mobile and other terminals.

The performance of competing data reduction algorithms for a mobile application was assessed using conventional fixed network telephone handsets. The opinion scores from the subjective experiment were then used to select the codec which performed best over the required range of noise and tandeming conditions. Later, the performance of the selected codec was subjectively judged in a mobile communications context, resulting in significantly higher subjective ratings. The reason for this variation was found to be that users of mobile equipment expect lower audio quality and had unconsciously adjusted their subjective ratings to match this expectation.

The objective perceptual analysis described above will provide an opinion prediction which does not include this user expectation effect. It is proposed that an absolute objective performance metric is established which can be accurately predicted by measurement algorithms. The expectation factor appropriate to a given product or market sector can then be explicitly applied.

3.2 *Cognitive aspects*

Moving into cognitive function and task dependency, simple rules which describe the subjectivity of error in controlled conditions become less relevant. For example, in an emergency the task of obtaining assistance would dominate the cognitive process and minor audible errors in the communication channel would be irrelevant. Similarly, generalised perceptual rules cannot account for an individual's state of mind (happy, sad, angry) at the time of the subjective judgement. It should be recalled that the variable nature of human subjective judgement is one of the motivations for developing an objective measurement system. If an 'average' listener's opinion of subjective audio quality can be predicted, this provides the required objective measurement allowing system performance to be compared and improved.

There are aspects of cognition which are consistent and which can be included in a simple rule-based model. The main example is the recognition of language where certain words and phrases are more critical to the cognitive task of recognition than others. Certain parts of language such as consonant onset (consonant plus second formant transition) are consistently critical to recognition and can be weighted more highly in an analysis of subjective audio quality. This is a practical proposition since the test-signal must be known in advance for this kind of *a priori* analysis. To date only this simple linguistic cognitive rule has been implemented in a practical system. There is clearly potential here for further development.

4. Applications for audio perceptual analysis

The main application areas for audio perceptual analysis are listed below:

- measurement tool
 - optimisation,
 - iterative design aid,
 - performance specification,
 - procurement selection,
 - refinement of proprietary equipment for service differentiation;
- measurement tool
 - speech quality assessment,
 - simple acceptance tester,
 - performance tester for commissioning and monitoring;
- perceptually motivated diagnostic tool, for speech and music
 - codec design,
 - expert fault finding,
 - psychoacoustic training aid;
- as part of a novel nonlinear network planning tool,
- analysis tool for opinion score prediction,
- research model, as a starting point for further development.

4.1 *Measurements tools*

An important requirement for many applications is the provision of a metric for speech quality which indicates that adequate performance is achieved, as perceived by the customer. When systems contain nonlinear processes this metric can best be provided by perceptually motivated analysis techniques. A prototype analysis based on a simplified perceptual model was developed to demonstrate a speech quality 'go/no-go' indicator for a mobile application.

When new equipment is installed it must be tested to ensure that performance is acceptable to the customer, and

monitored to ensure that performance remains acceptable, particularly for private networks where new equipment increasingly contains dynamic data reduction of speech channels. In these cases a measurement tool is required which can provide a repeatable, graduated measure of speech quality, providing the following important functions:

- supplying commissioning engineers with easily interpretable evidence that systems are performing correctly — the objective assessment of multiple channels will be quicker than a combination of informal listening and conventional measurement,
- establishing a reference for speech-quality performance which can be monitored over a period of time, and in particular can be used to demonstrate to a customer that an agreed level of performance is maintained, perhaps as part of a service level agreement (SLA) in terms of a subjectively relevant metric.

4.2 Nonlinear network planning tool

Existing linear network planning tools rely on performance parameters for individual network components being combined to provide a prediction of overall performance. When one or more of the network elements are significantly nonlinear it is no longer possible to determine adequate parameters which are additive in the same way.

A new concept for a network planning tool capable of predicting the performance on multiple nonlinear network components has been proposed. The planning tool concept is based on perceptual analysis + network emulation as shown in Fig 11.

The graphical user interface would allow the target network architecture to be specified as a block diagram. The controlling software would then appropriately configure the network emulation ready to be assessed by the perceptual analysis. The results showing both a prediction of subjective audio quality and a range of conventional transmission performance parameters would be displayed via the user interface.

As part of a recent feasibility study nonlinear network configurations were emulated, using the BT Network Emulator [20], and assessed with the perceptual analysis. The resulting prediction of their subjective performance was compared with that determined by subjective test with good correlation. While this does not represent an exhaustive feasibility study, which must consider the performance of a wide range of emulated network configurations, it does provide direct evidence of the feasibility of the new network planning tool concept.

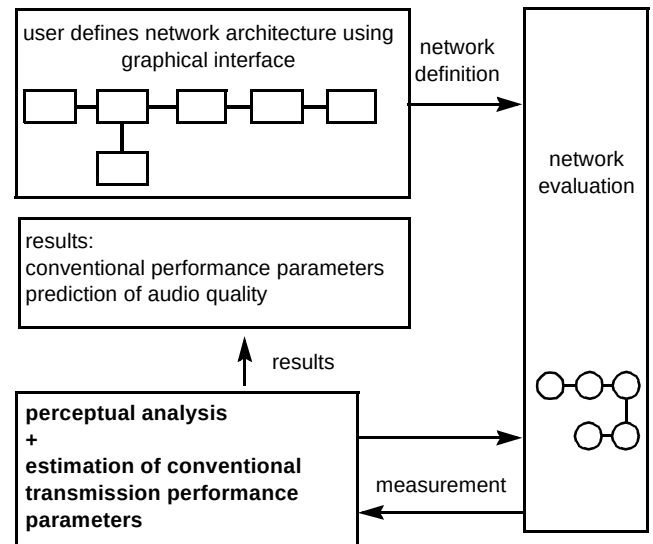


Fig 11 Planning tool concept.

4.3 Perceptual analysis to complement subjective experiment

Another important application for an established perceptual analysis will be to replace or complement subjective testing.

It is interesting to note that one of the earliest applications of the prototype auditory model was to help identify the test conditions for a large subjective test when doubt was cast on which condition subjects had actually experienced during a test. The prototype model was used in elementary fashion to show which sub-set of sentences had been subjected to particular degradations.

The main prerequisites for perceptual analysis as a replacement for subjective testing are:

- to compare many sets of subjective test results with perceptual analysis predictions in order to refine and verify its performance over a wide range of conditions and to establish the necessary criteria for an expectation factor, e.g. European Telecommunications Standardisation Institute competition for codec selection,
- to pilot (alpha trial) perceptual analysis in parallel with subjective tests in order to show that its predictions were satisfactory — this is very important to gain credibility and confidence in the system prior to its use in contributing to high-value commercial decisions.

Further work is required to process a variety of subjective test data sets through the model in order to establish its credibility sufficiently to replace subjective tests. Initially it is anticipated that the perceptual analysis could be used to complement a subjective test. The perceptual

analysis could objectively assess a set of test conditions with a sub-set of the conditions being subsequently confirmed by human subjects in a reduced subjective experiment. The objective verification of test conditions and the reduction in size of the subjective test together represent a substantial potential cost and time saving.

4.4 As a starting point for future research

The final application for the auditory perceptual analysis is to provide the starting point for future research into:

- wideband audio models,
- models for non-speech audio applications, e.g. music,
- the philosophical starting point for related work on visual perceptual models,
- objective multimedia assessment.

5. Multimedia assessment

Technology is opening up many new market opportunities and the way that we work, play and interact is changing [21]. Communications systems have historically been conducted in a single medium, but, with the advent of multimedia, this is now no longer the case. Multimedia, as the term suggests has the ability to stimulate the auditory, visual and tactile senses together. To address the perceptual understanding of a multimodal stimulus, a ‘cognitive’ framework based on multisensory perceptual analysis (see Fig 12) is starting to be developed. (In order to build this

‘cognitive’ model that can predict perception, when more than one of the senses are being stimulated, it is essential to understand many aspects of human behaviour previously not investigated.) People are good at identifying similar patterns, in pictures, sound, or human behaviour — but, how they do it remains somewhat of a mystery.

Objective assessment of multimedia products and services will require multisensory perceptual analysis. A visual perceptual model will be required to complement the audio model. Further, the relative contribution of each sense to the overall opinion of performance will depend on the task undertaken. A ‘cognitive’ layer model will therefore be required to provide the perceptual weightings relating to the task undertaken. Recent neurological evidence indicates that emotions are not a luxury — they are essential for ‘reason’ to function normally, even in rational decision-making. Furthermore, emotional expression is a natural and significant part of human interaction. Whether it is used to indicate like/dislike or interest/disinterest, emotion plays a key role in multimedia information retrieval and user preference modelling, and will form part of this ‘cognitive’ framework.

Only an elementary cognitive model is initially envisaged which appropriately weights the perceptual layer descriptors for a small number of well-defined task types, such as a videotelephony scenario.

The BT approach to perceptual modelling has an advantageous perceptual layer structure for this kind of sensory combination since the task weighting can be applied to one or more of the error descriptors for each sensory model.

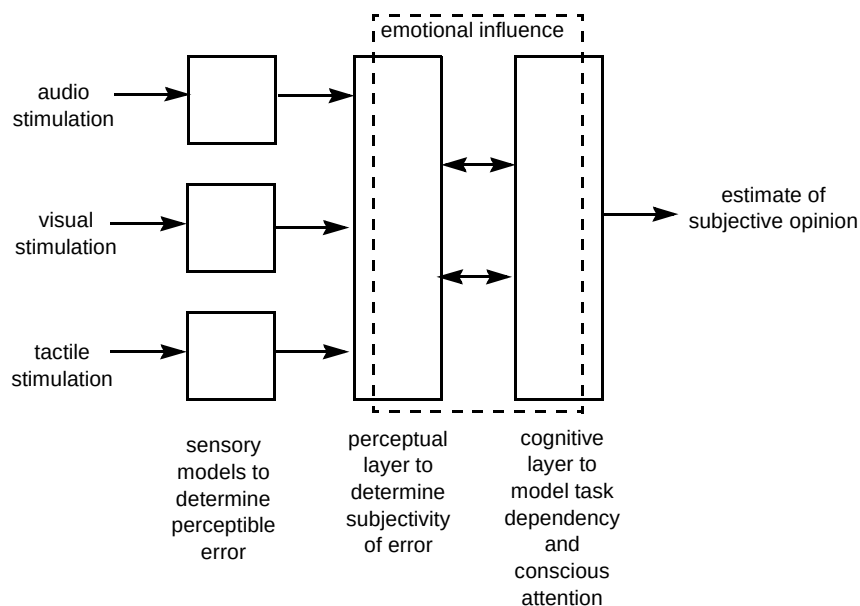


Fig 12 Concept for multisensory perceptual analysis.

6. Conclusions

The economic necessity of providing adequate subjective quality at a given price has been described together with the increasing presence of nonlinear processes such as data reduction in practical communications networks. Data reduction is already widespread in telecommunications, is appearing in broadcasting (e.g. digital audio broadcasting and MPEG) and professional audio (e.g. musicam), and has emerged in domestic electronics in both Digital Compact Cassette and MiniDisc. The need for a new generation of objective measures to provide adequate characterisation of such systems is apparent.

Objective performance assessment using an analysis based on a model of the human senses offers a potentially powerful and robust means of characterising the perceived performance of nonlinear systems. An audio perceptual analysis has been developed and was introduced together with a description of its main applications:

- speech quality assessment,
- perceptually motivated diagnostic tool,
- complement to subjective tests,
- a component of a novel nonlinear network planning tool,
- a starting point and enabler for further research.

The potential extension of perceptual analysis to include other human senses is also described. This is expected to be a key area for the communications industry since the bandwidth requirements for multimedia and virtual-reality applications are substantial and certain to contain data limitations or data reduction. It is argued that the perceptual implications of this data reduction in multisensory systems will require a perception-based performance analysis in order to provide a meaningful objective measure for:

- efficient optimisation of systems,
- specification and monitoring of a minimum perceived performance.

The algorithmic evaluation of perceived performance of a multisensory task must be related to the task undertaken

by the user. An outline model for a multisensory perceptual analysis is proposed including a cognitive layer which will initially weight the combination of the senses according to a limited range of task scenarios.

The ultimate goal of this work is to view information, not in terms of Shannon's notation, but as it is described and understood by the humans who perceive it.

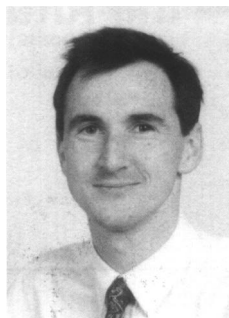
Acknowledgements

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With a background in audio engineering, he has worked on a number of telephony acoustics projects including the development of a novel noise cancelling hand-set. Since 1990 his work has included the development of objective measurement methods capable of predicting the performance of nonlinear audio systems. He received his PhD from the University of Essex in Colchester, in July 1995 in respect of his work on objective audio quality assessment and is currently extending his research into 3-D audio and multimedia assessment.

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Graham Cosier joined BT Laboratories as an apprentice in 1970. He successfully completed the CEI pt.11 examination in Electrical and Electronic Engineering, Computing and Control in 1980. During the period to 1984 he worked in the area of electro-acoustics and published a number of papers on auditory human perception. During that period he also contributed to the British Standard for hands-free telephony. In 1986 he became involved in digital mobile radio and was elected chairman of one of the ETSI technical groups for GSM. In 1988 he successfully completed the Diploma in

Management studies and in 1994 he was appointed UK co-ordinator for ITU SG12. He also represents BT on the Economic and Social Research Council. Since 1993 he has managed the Advanced Perception Unit within AA&T, looking into future telepresence applications.